

02 - Bayesian Decision Making

Pattern Recognition and Machine Learning, Lab 2

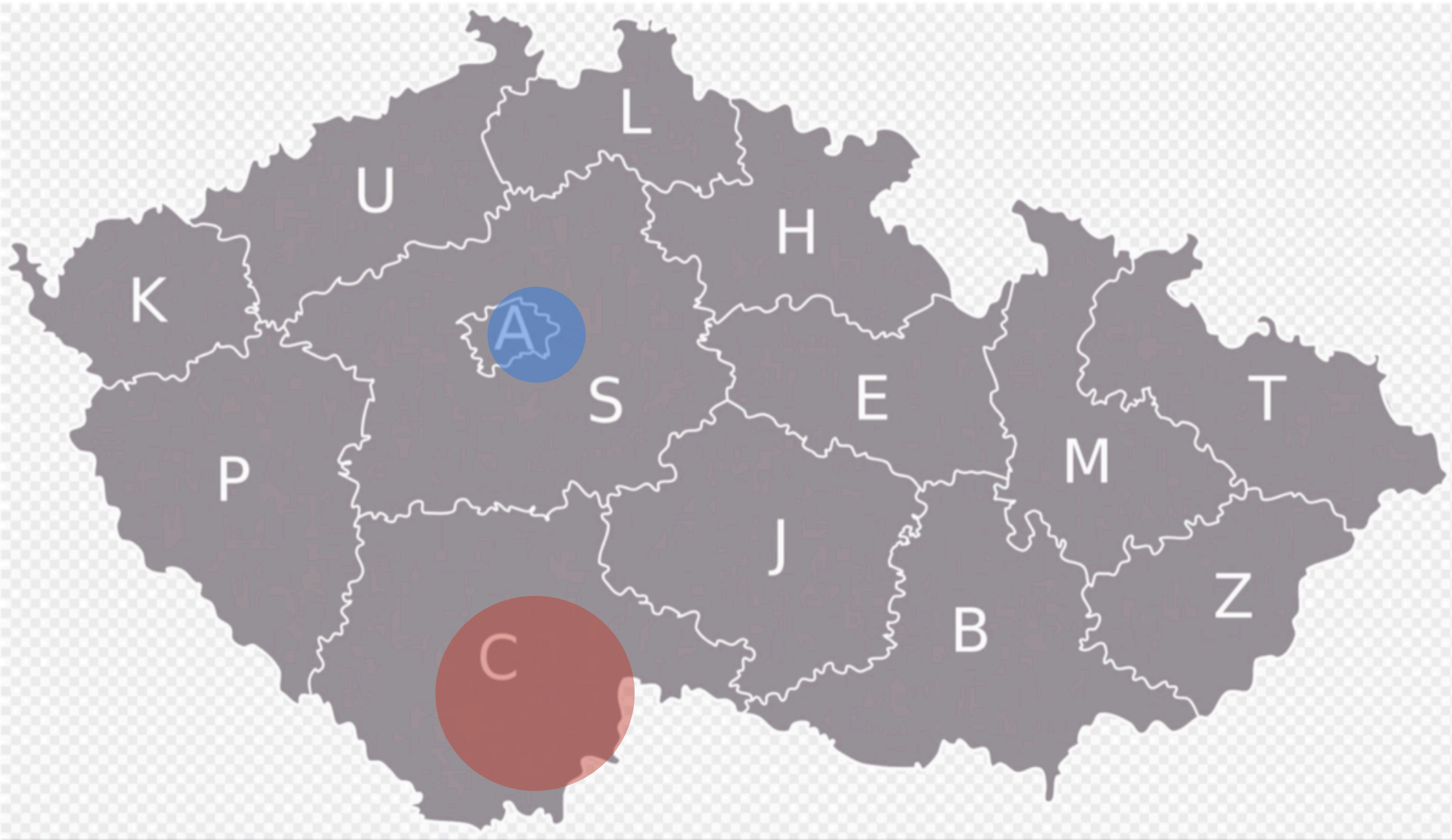
Michal Neoral

neoramic@fel.cvut.cz

2 October 2026

Motivation: licence plates

Kód	Kraj	2001–2004	Od 2004 do současnosti	Zatím nejvyšší řada ^[9]
A	Praha	1A7 4551	2AD 3824	8AJ
B	Jihomoravský kraj		6B2 6635	2BN
C	Jihočeský kraj	1C2 6377	7C2 5025	9C2
E	Pardubický kraj		2E7 7010	6E7
H	Královéhradecký kraj	2H2 7149	2H4 2270	7H8
J	Kraj Vysočina	2J2 6382	3J1 2272	6J9
K	Karlovarský kraj		3K3 2886	4K8
L	Liberecký kraj		2L0 2404	6L1
M	Olomoucký kraj	1M4 0000	6M1 3246	7M3
P	Plzeňský kraj	1P5 0858	7P2 4159	8P4
S	Středočeský kraj	3S9 6552	7S3 3237	5SK
T	Moravskoslezský kraj		6T9 9994	1TZ
U	Ústecký kraj		6U3 2183	1UH
Z	Zlínský kraj		4Z0 7472	6Z9

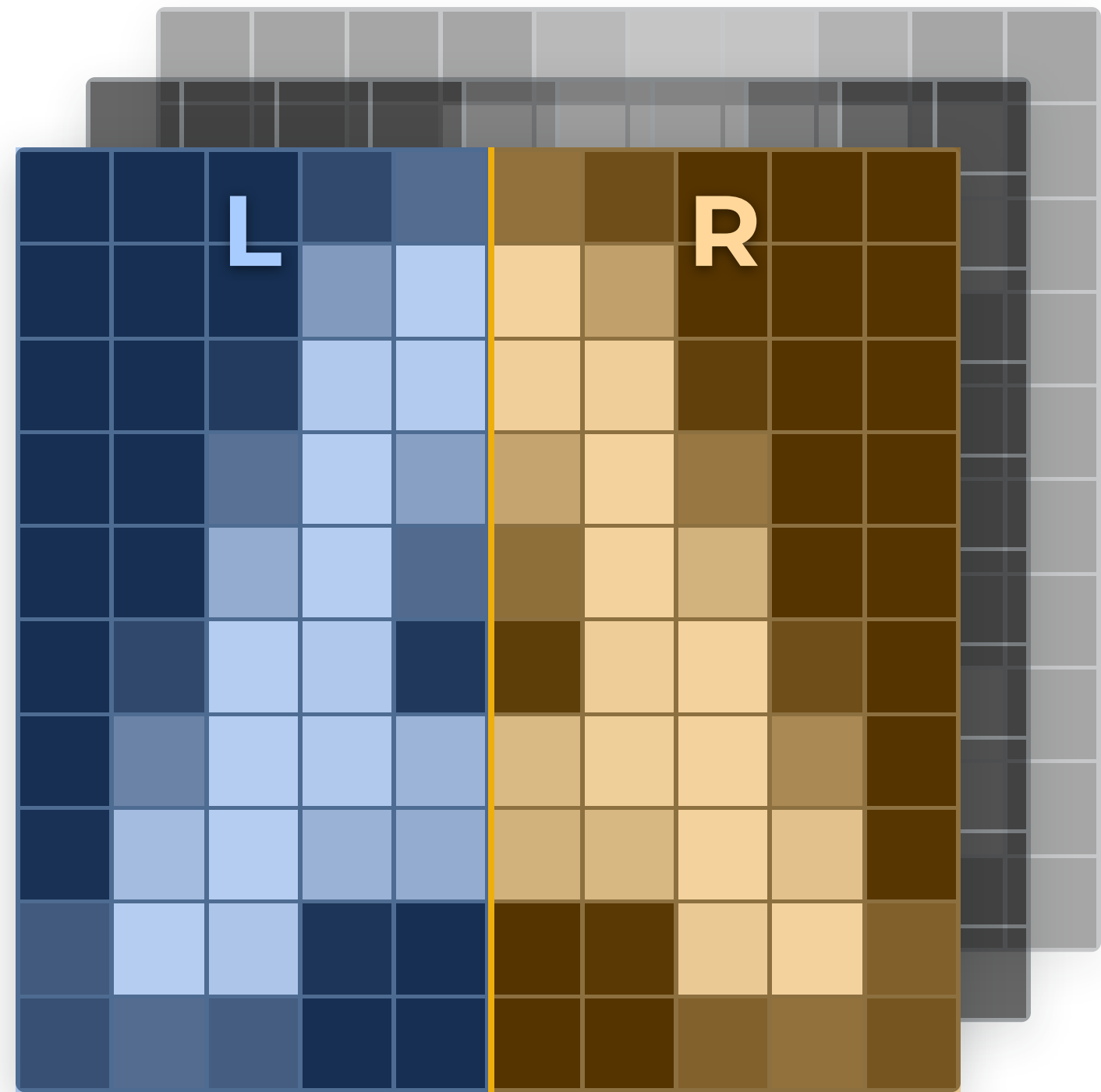


The second letter of an older plate tells the **region** (kraj).



Our simplification: recognize only the letters **A** (Praha) and **C** (Jihočeský kraj).

The data and the features



N images of 10 × 10 pixels



How do we get a (discrete) feature x from an image of a letter?

L = sum of the pixel values in the **left** half

R = sum of the pixel values in the **right** half

$$x = g(L - R) \in X = \{-10, \dots, 10\}$$

g quantizes the difference to a small set of integers

The Bayesian risk

? How do we define the Bayesian risk $R(q)$ of a strategy q ?

$$R(q) = \sum_{x \in X} \sum_{k \in K} p_{XK}(x, k) W(k, q(x))$$

Symbol	Meaning	Licence plates	Mushrooms
X	observations $x \in X$: what we know	$x = g(L - R)$	cap colour, smell, ring
K	hidden states $k \in K$: what we don't know	letters $\{A, C\}$	edible, poisonous
D	decisions $d \in D$	$D = K = \{A, C\}$	eat, leave it ($D \neq K$)
p_{XK}	joint probability $p_{XK}(x, k)$	feature x with letter k	a red smelly one is poisonous
W	loss $W(k, d)$: price of d when the truth is k	0/1 loss	poisonous eaten: hospital
q	strategy $q : X \rightarrow D$	A or C for every x	"red with dots: leave it"

? What is the Bayesian (optimal) strategy q^* ?

What do we know?

Known

- $p(x \mid k)$: likelihood
- $p(k)$: prior

Unknown

- $p_{XK}(x, k)$: joint probability
- it appears in the risk $R(q)$

? How do we get $p_{XK}(x, k)$?

Derivation: the joint probability

Product rule $p_{XK}(x, k) = p(x | k) p(k) = p(k | x) p(x)$

Marginal $p(x) = \sum_{k \in K} p_{XK}(x, k) = \sum_{k \in K} p(x | k) p(k)$

Bayes' rule $p(k | x) = \frac{p(x | k) p(k)}{p(x)}$

The risk now contains only known quantities:

$$R(q) = \sum_{x \in X} \sum_{k \in K} p(x | k) p(k) W(k, q(x))$$

Discrete example

Bayes example: discrete x

$x = g(L - R) \in X = \{x_1, x_2, x_3, x_4\}, \quad K = D = \{A, C\}$

Priors: $p(A) = p(C) = 0.5$

Likelihoods $p(x \mid k)$

	x_1	x_2	x_3	x_4
A	0.3	0.2	0.1	0.4
C	0.7	0.2	0.1	0

Loss $W(k, d)$

$k \backslash d$	A	C
A	0	1
C	1	0

Strategy $q(x)$

x_1	x_2	x_3	x_4
A	A	C	C



1. What is the risk $R(q)$ of this strategy?



2. How do we find the optimal strategy q^* ?



3. How do we determine the classification error?

Derivation: risk of the given strategy

$$R(q) = \sum_{x \in X} \sum_{k \in K} p(x \mid k) p(k) W(k, q(x))$$

With the 0/1 loss, only the wrong classes $k \neq q(x)$ contribute:

$$R(q) = \sum_{x \in X} \sum_{k \neq q(x)} p(x \mid k) p(k)$$

	x_1	x_2	x_3	x_4
decision $q(x)$	A	A	C	C
wrong class k	C	C	A	A
$p(x \mid k) p(k)$	$0.7 \cdot 0.5 = 0.35$	$0.2 \cdot 0.5 = 0.1$	$0.1 \cdot 0.5 = 0.05$	$0.4 \cdot 0.5 = 0.2$

$$R(q) = 0.35 + 0.1 + 0.05 + 0.2 = \mathbf{0.7}$$

Derivation: the optimal strategy

$$R(\boldsymbol{q}) = \sum_{\boldsymbol{x} \in \boldsymbol{X}} \underbrace{\sum_{\boldsymbol{k} \in \boldsymbol{K}} p(\boldsymbol{x} \mid \boldsymbol{k}) p(\boldsymbol{k}) W(\boldsymbol{k}, \boldsymbol{q}(\boldsymbol{x}))}_{R(\boldsymbol{x}, \boldsymbol{q}(\boldsymbol{x})): \text{partial risk}}$$

Each term depends only on the single decision $\boldsymbol{q}(\boldsymbol{x})$, so we minimize every term separately:

$$\min_{\boldsymbol{q}} R(\boldsymbol{q}) = \sum_{\boldsymbol{x} \in \boldsymbol{X}} \min_{\boldsymbol{d} \in \boldsymbol{D}} R(\boldsymbol{x}, \boldsymbol{d}), \quad \boldsymbol{q}^*(\boldsymbol{x}) = \arg \min_{\boldsymbol{d} \in \boldsymbol{D}} R(\boldsymbol{x}, \boldsymbol{d})$$

For the 0/1 loss and two classes:

$$R(\boldsymbol{x}, \boldsymbol{A}) = p(\boldsymbol{x} \mid \boldsymbol{C}) p(\boldsymbol{C}), \quad R(\boldsymbol{x}, \boldsymbol{C}) = p(\boldsymbol{x} \mid \boldsymbol{A}) p(\boldsymbol{A})$$

Example for x_1 :

$$R(x_1, \boldsymbol{A}) = 0.7 \cdot 0.5 = 0.35, \quad R(x_1, \boldsymbol{C}) = 0.3 \cdot 0.5 = \mathbf{0.15}, \quad \boldsymbol{q}^*(x_1) = \boldsymbol{C}$$

Discrete example: results

1. Risk of the given strategy

$$R(q) = p(A) (0.1 + 0.4) + p(C) (0.7 + 0.2) = \mathbf{0.7}$$

3. Classification error

$$\text{error} = \frac{\# \text{ misclassified}}{\# \text{ all} = N}$$

2. Partial risk and the optimal strategy

$$R(x, d) = \sum_{k \in K} p(x \mid k) p(k) W(k, d), \quad q^*(x) = \arg \min_{d \in D} R(x, d)$$

	x_1	x_2	x_3	x_4
$R(x, A)$	0.35	0.1	0.05	0
$R(x, C)$	0.15	0.1	0.05	0.2
$q^*(x)$	C	A or C	A or C	A

$R(q^*) = 0.15 + 0.1 + 0.05 + 0 = \mathbf{0.3}$. For x_2, x_3 both decisions are optimal (a tie), a preference must be agreed on.

Continuous x

Bayes example: continuous x

$x = L - R \in X = \mathbb{R}$ is now **continuous**

Likelihood: normal distribution

$$p(x \mid k) = \frac{1}{\sigma_k \sqrt{2\pi}} \exp\left(-\frac{(x - \mu_k)^2}{2\sigma_k^2}\right)$$

Given:

$\mu_A, \mu_C, \sigma_A, \sigma_C, p(A), p(C), W = 0/1$ loss

? How does the Bayesian risk $R(q)$ change for a continuous x ?

? What is the optimal strategy $q^*(x)$?

Derivation: optimal strategy, continuous x

$$R(q) = \int_X \sum_{k \in K} p_{XK}(x, k) W(k, q(x)) dx = \int_X p(x) \sum_{k \in K} p(k | x) W(k, q(x)) dx$$

$p(x) \geq 0$, so we minimize the integrand separately for every x :

$$q^*(x) = \arg \min_{d \in D} \sum_{k \in K} p(k | x) W(k, d)$$

For the 0/1 loss: $\sum_{k \neq d} p(k | x) = 1 - p(d | x)$, therefore

$$q^*(x) = \arg \max_{d \in D} p(d | x) = \arg \max_{d \in D} \frac{p(x | d) p(d)}{p(x)} = \arg \max_{d \in D} p(x | d) p(d)$$

Two classes: a quadratic inequality

We decide A where $p(x | A) p(A) \geq p(x | C) p(C)$:

$$p(A) \frac{1}{\sigma_A \sqrt{2\pi}} e^{-\frac{(x-\mu_A)^2}{2\sigma_A^2}} \geq p(C) \frac{1}{\sigma_C \sqrt{2\pi}} e^{-\frac{(x-\mu_C)^2}{2\sigma_C^2}}$$

$\sqrt{2\pi}$ cancels, both sides are positive, so we take the logarithm:

$$\ln p(A) - \ln \sigma_A - \frac{(x - \mu_A)^2}{2\sigma_A^2} \geq \ln p(C) - \ln \sigma_C - \frac{(x - \mu_C)^2}{2\sigma_C^2}$$

Move the terms without x to the right:

$$-\frac{(x - \mu_A)^2}{2\sigma_A^2} + \frac{(x - \mu_C)^2}{2\sigma_C^2} \geq \ln \left(\frac{\sigma_A}{\sigma_C} \cdot \frac{p(C)}{p(A)} \right) =: L$$

Two classes: collecting the powers of x

$$-\frac{(x - \mu_A)^2}{2\sigma_A^2} + \frac{(x - \mu_C)^2}{2\sigma_C^2} \geq L \quad \Big| \cdot 2\sigma_A^2\sigma_C^2 > 0$$
$$-\sigma_C^2(x - \mu_A)^2 + \sigma_A^2(x - \mu_C)^2 \geq 2\sigma_A^2\sigma_C^2L$$

Expand the squares:

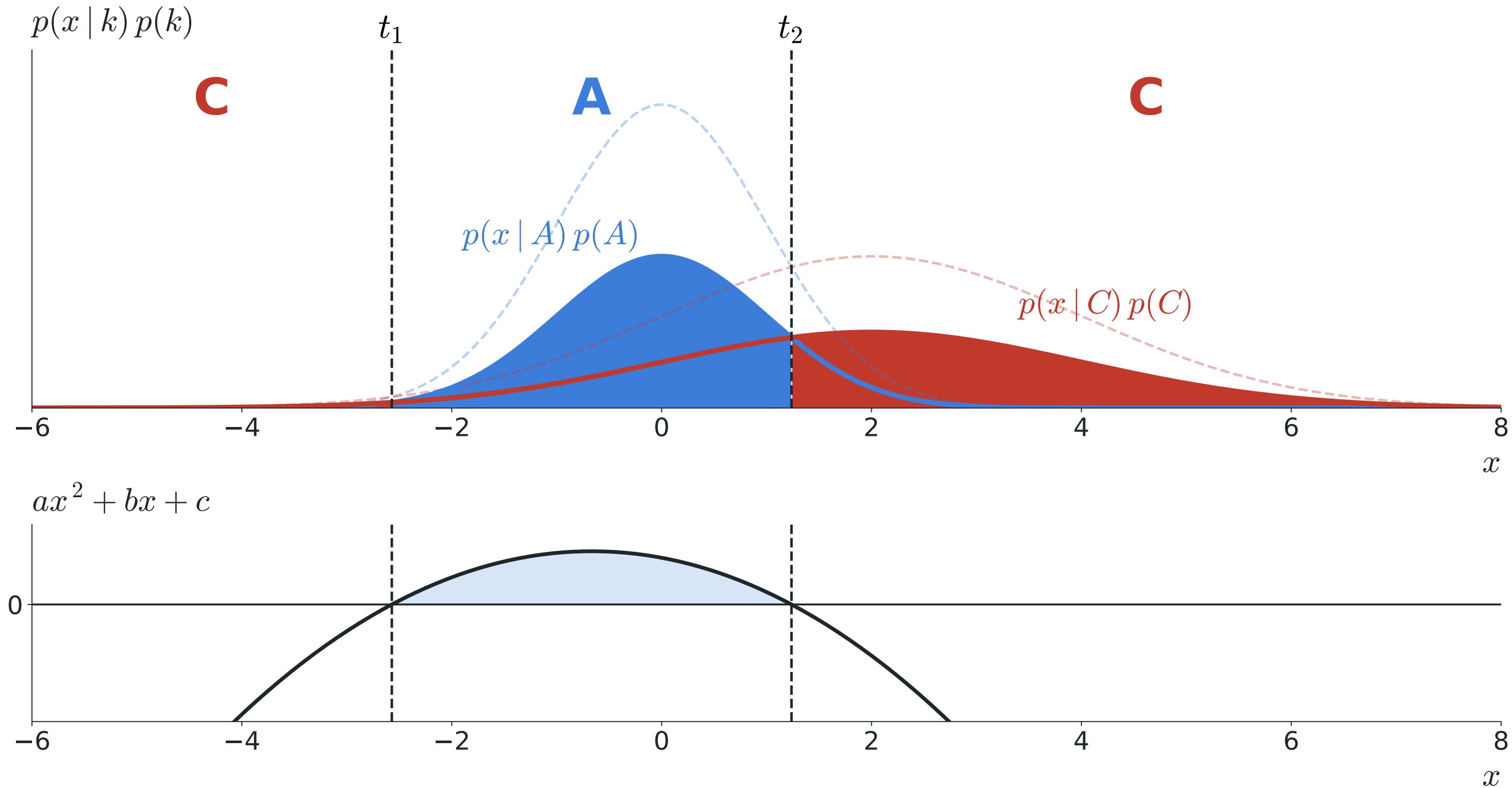
$$-\sigma_C^2x^2 + 2\sigma_C^2\mu_Ax - \sigma_C^2\mu_A^2 + \sigma_A^2x^2 - 2\sigma_A^2\mu_Cx + \sigma_A^2\mu_C^2 - 2\sigma_A^2\sigma_C^2L \geq 0$$

Group by the powers of x :

$$(\sigma_A^2 - \sigma_C^2)x^2 + 2(\sigma_C^2\mu_A - \sigma_A^2\mu_C)x + \sigma_A^2\mu_C^2 - \sigma_C^2\mu_A^2 - 2\sigma_A^2\sigma_C^2L \geq 0$$

$$a = \sigma_A^2 - \sigma_C^2 \quad b = 2(\sigma_C^2\mu_A - \sigma_A^2\mu_C) \quad c = \sigma_A^2\mu_C^2 - \sigma_C^2\mu_A^2 - 2\sigma_A^2\sigma_C^2L$$

Decision regions



The thresholds are the roots of $ax^2 + bx + c = 0$, class **A** where it is ≥ 0

Decision regions: try it

μ_A

0

σ_A

1

μ_C

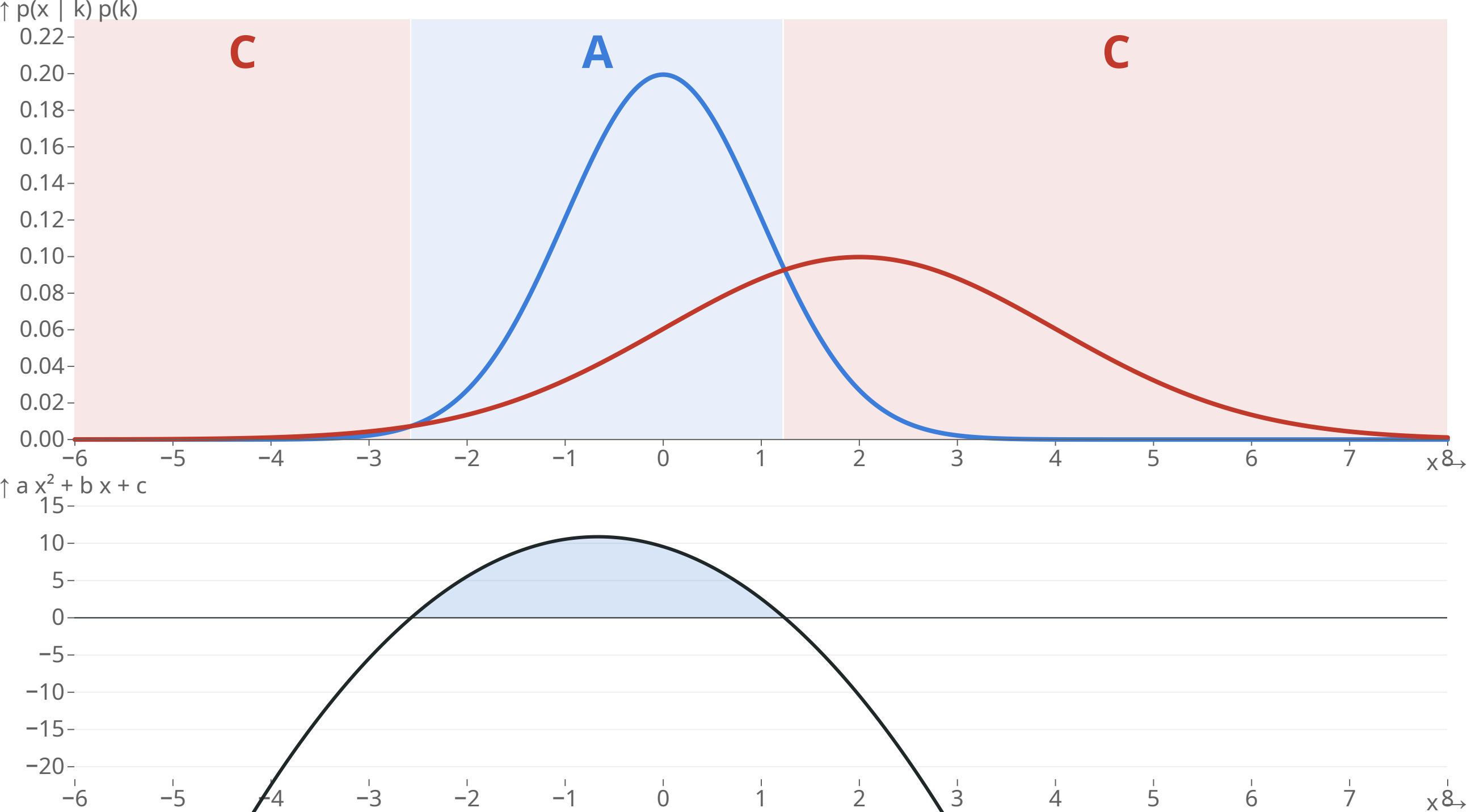
2

σ_C

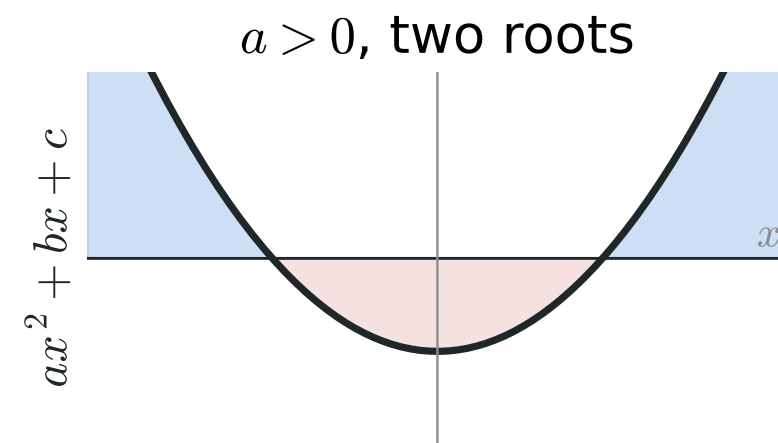
2

$p(A)$

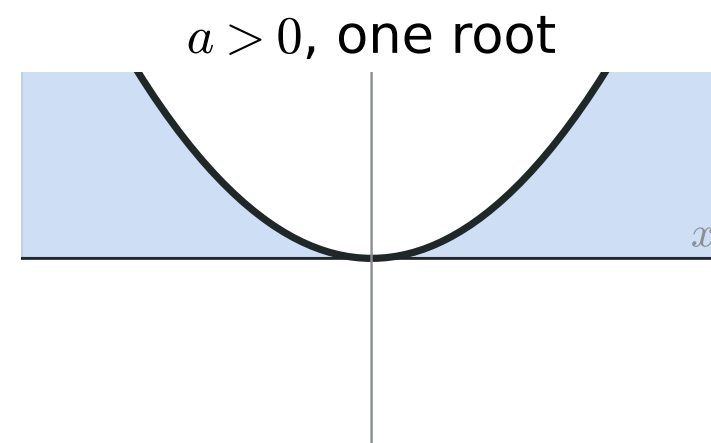
0.5



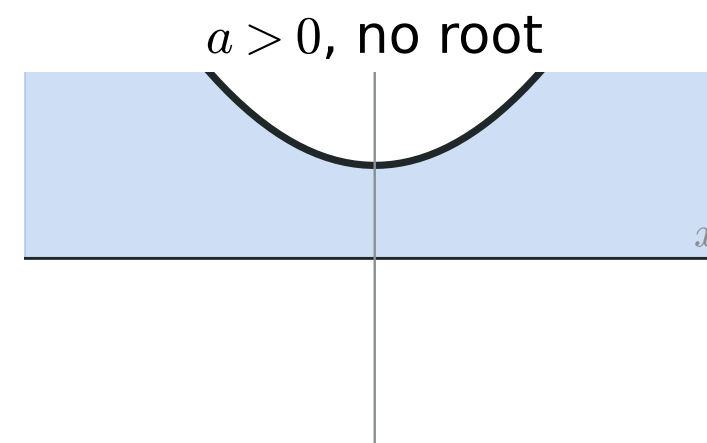
Watch out: all the cases of $ax^2 + bx + c \geq 0$



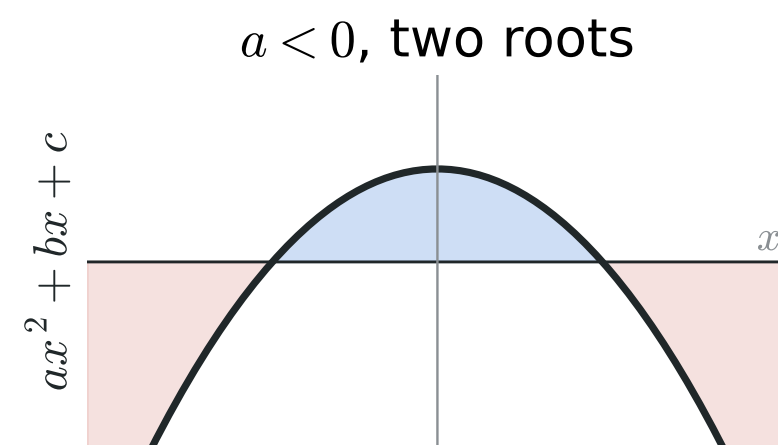
A | C | A



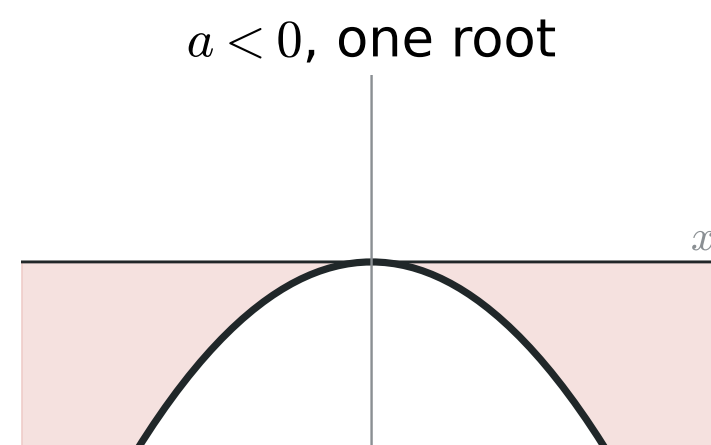
A everywhere



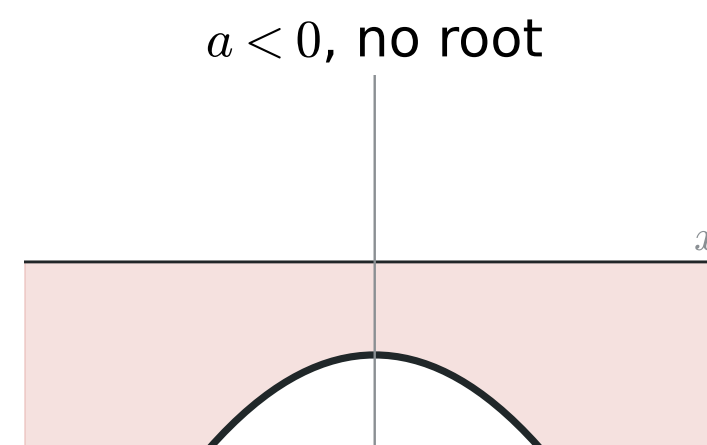
A everywhere



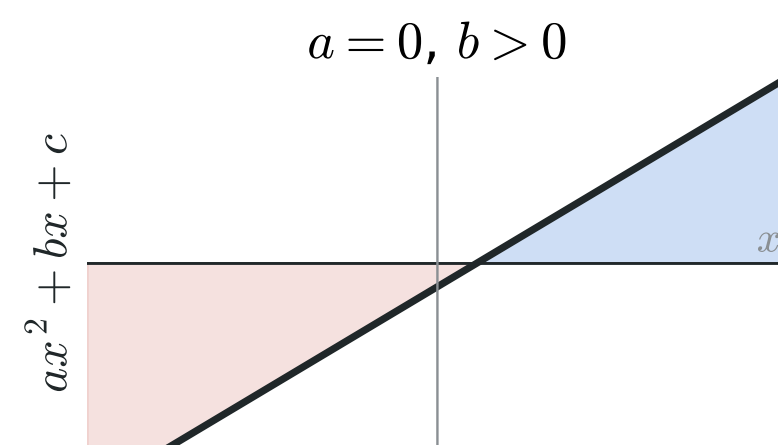
C | A | C



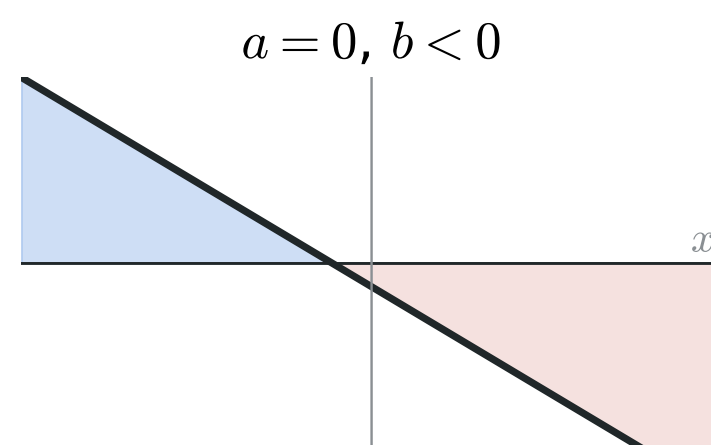
C (A in one point)



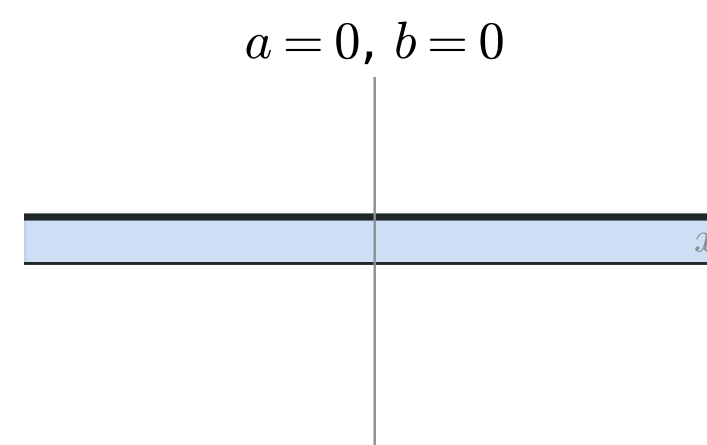
C everywhere



C | A



A | C



A if $c \geq 0$, else C



Which intervals
belong to class **A** in
each case?



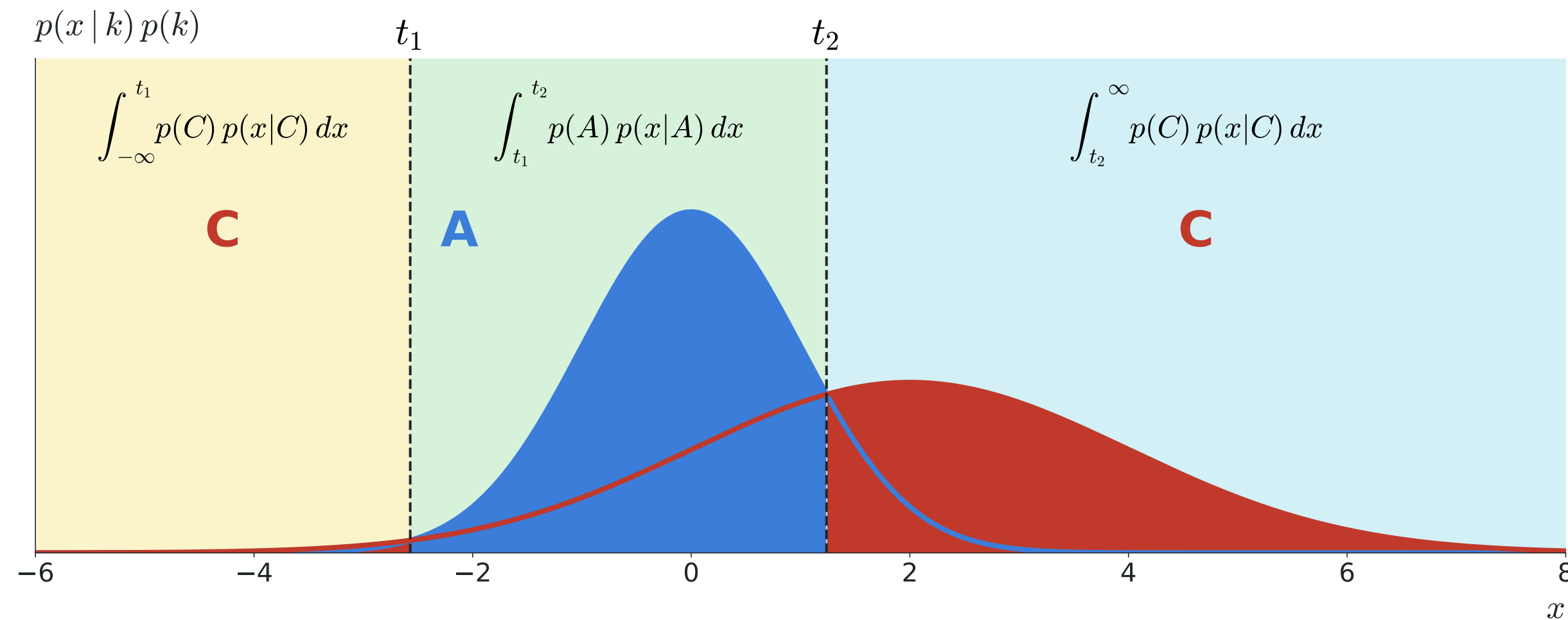
Handle **every case** in
your code, including
 $a = 0$ and no
real roots.

Output of the assignment



The strategy is stored as two thresholds $q['t1'] \leq q['t2']$ and the decisions for the three intervals, e.g. $q['decision'] = ['C', 'A', 'C']$.

The risk for continuous x



Compute the risk $R(\textcolor{purple}{q})$ of your strategy.



Hint: the area under a normal density up to t is `scipy.stats.norm.cdf(t, mu, sigma)`

$$R(\textcolor{purple}{q}) = 1 - \left(\int_{-\infty}^{t_1} \textcolor{green}{p}(\textcolor{red}{C}) \textcolor{teal}{p}(\textcolor{teal}{x}|\textcolor{red}{C}) dx + \int_{t_1}^{t_2} \textcolor{green}{p}(\textcolor{blue}{A}) \textcolor{teal}{p}(\textcolor{teal}{x}|\textcolor{blue}{A}) dx + \int_{t_2}^{\infty} \textcolor{green}{p}(\textcolor{red}{C}) \textcolor{teal}{p}(\textcolor{teal}{x}|\textcolor{red}{C}) dx \right)$$

Derivation: the risk for continuous x

$$R(q) = \int_{-\infty}^{\infty} \sum_{k \in K} p(k) p(x | k) W(k, q(x)) dx$$

0/1 loss: all classes except the decided one $q(x)$:

$$R(q) = \int_{-\infty}^{\infty} \left(\sum_{k \in K} p(k) p(x | k) - p(q(x)) p(x | q(x)) \right) dx$$

The first part integrates to one:

$$\int_{-\infty}^{\infty} \sum_k p(k) p(x | k) dx = \sum_k p(k) \int_{-\infty}^{\infty} p(x | k) dx = \sum_k p(k) = 1$$

$$R(q) = 1 - \int_{-\infty}^{\infty} p(q(x)) p(x | q(x)) dx$$

The integral is the probability of a correct decision.

Derivation: splitting into intervals

q is constant on the intervals (t_{i-1}, t_i) with decisions d_1, d_2, d_3 , where $t_0 = -\infty, t_3 = \infty$:

$$R(q) = 1 - \sum_{i=1}^3 p(d_i) \int_{t_{i-1}}^{t_i} p(x | d_i) dx$$

Cumulative distribution function of class k , in code `norm.cdf(t, mu, sigma)`:

$$F_k(t) = \int_{-\infty}^t p(x | k) dx, \quad \int_a^b p(x | k) dx = F_k(b) - F_k(a)$$

with $F_k(-\infty) = 0$ and $F_k(\infty) = 1$.

For the decisions $C | A | C$:

$$R(q) = 1 - \left(p(C) F_C(t_1) + p(A) (F_A(t_2) - F_A(t_1)) + p(C) (1 - F_C(t_2)) \right)$$

Derivation: a numerical example

Parameters:

$$\mu_A = 0, \sigma_A = 1, \mu_C = 2, \sigma_C = 2, p(A) = 0.5$$

Thresholds:

$$t_1 = -2.571, t_2 = 1.238, \text{decisions } C \mid A \mid C$$

$$F_C(t_1) = 0.0111, \quad F_A(t_1) = 0.0051, \quad F_A(t_2) = 0.8921, \quad F_C(t_2) = 0.3515$$

$$R(q) = 1 - 0.5 (0.0111 + (0.8921 - 0.0051) + (1 - 0.3515)) = 1 - 0.7733 = \mathbf{0.2267}$$

Check: the areas of the wrong class give the same result:

$$R(q) = p(A)(F_A(t_1) + 1 - F_A(t_2)) + p(C)(F_C(t_2) - F_C(t_1)) = 0.2267$$

Take-away for the lab

- ✓ **Discrete x :** risk of a given strategy, the optimal strategy, classification error
- ✓ **Continuous x :** coefficients a, b, c of the quadratic inequality
- ✓ Thresholds t_1, t_2 and the **decisions** for the three intervals
- ✓ **Risk** of the strategy using `scipy.stats.norm.cdf`
- ✓ Watch out for **all the cases** ($a = 0$, one or no real root)

